

# Elements of Statistics and Probability

Nabil Ahmed Uthso

School of Data and Computational Sciences

BRAC University

E-mail: [nabil.ahmed@bracu.ac.bd](mailto:nabil.ahmed@bracu.ac.bd)

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## 1 Some Special Discrete Probability Distributions

- Binomial Distribution
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# Binomial Distribution

- **Bernoulli Trial:** A Bernoulli trial is a random experiment that satisfies the following conditions:
  - 1 There are only two possible outcomes for each trial, one may arbitrarily be called a success and other a failure
  - 2 The probability of success is the same for each trial. If the probability of success in a trial is  $p$ , it remains constant from trial to trial. Then the probability of failure is  $1 - p = q$
  - 3 The trials can be repeated  $n$  times, where  $n$  is a constant
  - 4 The trials are independent

# Binomial Distribution

- Let  $X$  be a discrete r.v. and  $f(x)$  be the pmf of  $X$ . If  $f(x)$  is given by

$$f(x) = \binom{n}{x} p^x q^{n-x}; \quad x = 0, 1, 2, \dots, n, \quad 0 < p < 1, \quad q = 1 - p$$

- then  $X$  is said to have a binomial distribution
- Therefore,

$$P(X = 0) = f(0) = q^n$$

$$P(X = 1) = f(1) = npq^{n-1}$$

...

$$P(X = n) = f(n) = p^n$$

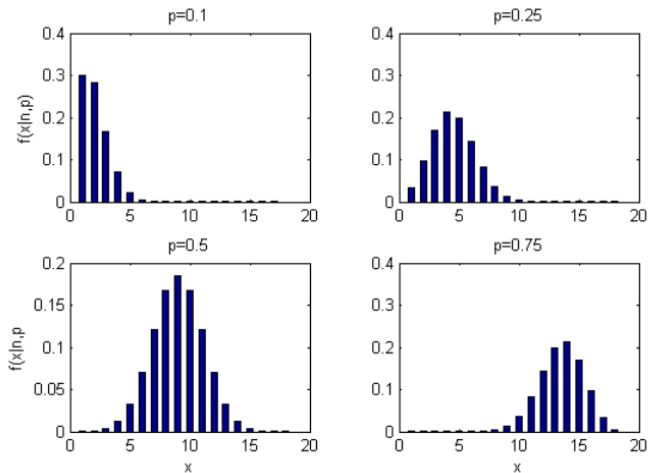
# Bionomial Distribution

- Show that binomial pmf is a valid pmf:

$$\begin{aligned}\sum_{x=0}^n f(x) &= \sum_{x=0}^n \binom{n}{x} p^x q^{n-x} \\&= q^n + npq^{n-1} + \dots + p^n \\&= (q + p)^n \\&= 1^n; [\text{since } 1 - p = q] \\&= 1\end{aligned}$$

- Therefore, it is a probability function

# Shape of a Binomial Distribution



# Example

- Suppose 4% of BracU graduate do not get any job. From a sample of 20 graduates, what is the probability that
  - (a) exactly 5 of them are unemployed.
  - (b) none of them is unemployed.
  - (c) two or fewer are unemployed.
  - (d) maximum 4 of them are unemployed.
  - (e) at least 3 of them are unemployed.

# Example

- **Solution:** Here,  $n = 20, p = 0.04$

$$(a) P(X = 5) = \binom{20}{5} 0.04^5 (1 - 0.04)^{20-5} = 0.00009$$

$$(b) P(X = 0) = \binom{20}{0} (0.04)^0 (1 - 0.04)^{20-0} = 0.442$$

$$(c) P(X \leq 2) = \sum_{x=0}^2 \binom{20}{x} (0.04)^x (1 - 0.04)^{20-x} = 0.956$$

$$(d) P(X \leq 4) = \sum_{x=0}^4 \binom{20}{x} (0.04)^x (1 - 0.04)^{20-x} = 0.999$$

$$(e) P(X \geq 3) = 1 - P(X \leq 2) = 1 - 0.956 = 0.044$$

# Mean and Variance of a Binomial R.V.

- Mean:

$$\begin{aligned} E(X) &= \sum xf(x) \\ &= \sum_{x=0}^n x \binom{n}{x} p^x q^{n-x} \\ &= \sum_{x=0}^n x \frac{n!}{x!(n-x)!} p^x q^{n-x} \\ &= \sum_{x=0}^n x \frac{n(n-1)!}{x(x-1)!(n-x)!} p p^{n-1} q^{n-x} \\ &= np \sum_{x=1}^n \binom{n-1}{x-1} p^{x-1} q^{n-x} \\ &= np \times 1 \\ &= np \end{aligned}$$

# Mean and Variance of a Binomial R.V.

- Variance:

- We know that  $V(X) = E(X^2) - [E(X)]^2$
- Observe that  $x^2 = x(x-1) + x$
- Therefore,  $E(X^2) = E[X(X-1)] + E(X)$

$$\begin{aligned}E[X(X-1)] &= \sum x(x-1) \binom{n}{x} p^x q^{n-x} \\&= \sum x(x-1) \frac{n(n-1)(n-2)!}{x(x-1)(x-2)!(n-x)!} p^x q^{n-x} \\&= n(n-1)p^2 \sum_{x=2}^n \binom{n-2}{x-2} p^{x-2} q^{n-x} \\&= n(n-1)p^2 \times 1 \\&= n(n-1)p^2\end{aligned}$$

# Mean and Variance of a Binomial R.V.

- Therefore,  $E(X^2) = n(n-1)p^2 + np$
- So, the variance will be

$$\begin{aligned}V(X) &= E(X^2) - [E(X)]^2 \\&= n(n-1)p^2 + np - (np)^2 \\&= n^2p^2 - np^2 + np - n^2p^2 \\&= np - np^2 \\&= np(1-p) \\&= npq\end{aligned}$$

# Exercise

- (1) Let  $X \sim b(n, p)$ . Show that  $E\left(\frac{X}{n}\right) = p$  and  $E\left(\frac{X}{n} - p\right)^2 = \frac{pq}{n}$
- (2) Prove the following:
- (a)  $b(x; n, p) = b(n - x; n, 1 - p)$
  - (b)  $b(x; n, p) = F(x; n, p) - F(x - 1; n, p)$
- (3) Let  $X \sim b(2, p)$  and  $Y \sim b(4, p)$ . If  $P(X \geq 1) = \frac{5}{9}$ , then  $P(Y \geq 1) = ?$

## Exercise 1 (binomial)

Based on data collected by the National Center for Health Statistics and made available to the public in the Sample Adult database (A-5), an estimate of the percentage of adults who have at some point in their life been told they have hypertension is 23.53 percent. If we select a simple random sample of 20 U.S. adults and assume that the probability that each has been told that he or she has hypertension is .24, find the probability that the number of people in the sample who have been told that they have hypertension will be:

- (a) Exactly three
- (b) Three or more
- (c) Fewer than three
- (d) Between three and seven, inclusive
- (e) How many adults who have been told that they have hypertension would you expect to find in a sample of 20?

# Solution to Exercise 1 (Binomial)

Here, number of trials,  $n = 20$ , probability,  $p = 0.24$

(a)  $P(X = 3) = P(X \leq 3) - P(X \leq 2)$

(b)  $P(X \geq 3) = 1 - P(X \leq 2)$

(c)  $P(X \leq 3) = P(X \leq 2)$

(d)  $P(3 \leq X \leq 7) = P(X \leq 7) - P(X \leq 3)$

(e)  $E(X) = n \times p = 20 \times 0.24 = 4.8 \approx 5$

## Exercise 2 (binomial)

The same survey mentioned in the previous exercise shows that 32 percent of U.S. adults indicated that they have been tested for HIV at some point in their life. Consider a simple random sample of 15 adults selected at that time. Find the probability that the number of adults who have been tested for HIV in the sample would be:

- (a) Three
- (b) Less than five
- (c) Between five and nine, inclusive
- (d) More than five, but less than 10
- (e) Six or more

## Exercise 3 (binomial)

Coughlin et al. (A-6) estimated the percentage of women living in border counties along the southern United States with Mexico (designated counties in California, Arizona, New Mexico, and Texas) who have less than a high school education to be 18.7. Assume the corresponding probability is .19. Suppose we select three women at random. Find the probability that the number with less than a high school education is:

- (a) Exactly zero
- (b) Exactly one
- (c) More than one
- (d) Two or fewer
- (e) Two or three
- (f) Exactly three

## Solution to Exercise 3 (Binomial)

Here, number of trials  $n = 3$ , probability,  $p = 0.19$

(a)  $P(X = 0)$

(b)  $P(X = 1) = P(X \leq 1) - P(X = 0)$

(c)  $P(X > 1) = 1 - P(X \leq 1)$

(d)  $P(X \leq 2)$

(e)  $P(X = 2) + P(X = 3)$

(f)  $P(X = 3)$

## Exercise 4 (binomial)

In a survey of nursing students pursuing a master's degree, 75 percent stated that they expect to be promoted to a higher position within one month after receiving the degree. If this percentage holds for the entire population, find, for a sample of 15, the probability that the number expecting a promotion within a month after receiving their degree is:

- (a) Six
- (b) At least seven
- (c) No more than five
- (d) Between six and nine, inclusive

# Rapid fire questions (binomial)

- (1) 4% of skilled people do not get a job. From 20 people, what is the probability of 2 or fewer unemployed workers?
- (2) In a question paper, there are 10 questions. Each has 4 possible answers. By guessing, what is the probability of correcting 3 to 5 questions?
- (3) A die is thrown repeatedly 36 times in all. Find  $E(X)$  and  $V(X)$ , where  $X$  is the number of sixes obtained.
- (4) A computer installation has 10 terminals. Independently, the probability that any one terminal will require attention during a week is 0.1. What is the probability that 2 terminals will require attention during the next week.

# Geometric Distribution

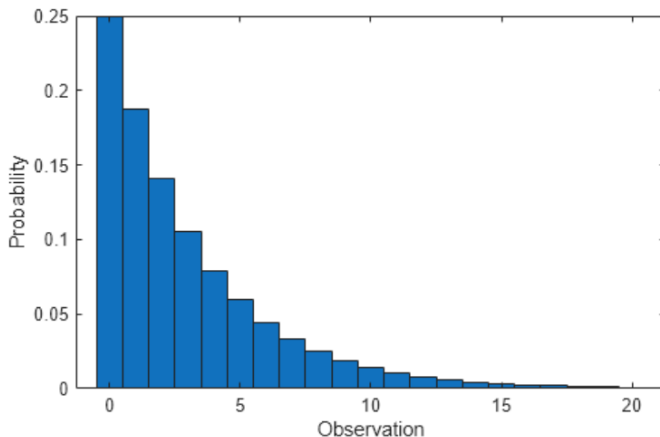
- A discrete r.v.  $X$  is said to have a geometric distribution if its pmf is given by

$$f(x) = pq^x; x = 0, 1, \dots, 0 < p < 1, q = 1 - p$$

- Here,  $X$  is the number of failure before the first success
- It is a valid pmf:

$$\begin{aligned}\sum_{x=0}^{\infty} f(x) &= \sum_{x=0}^{\infty} pq^x \\ &= p \sum_{x=0}^{\infty} q^x \\ &= p \times [1 + q + q^2 + \dots] \\ &= p \times \frac{1}{1 - q} \\ &= p \times \frac{1}{p} = 1\end{aligned}$$

# Shape of a Geometric Distribution



# Mean and Variance of a Geometric R.V.

- Mean:

$$\begin{aligned} E(X) &= \sum_{x=0}^{\infty} x p q^x \\ &= p q \sum_{x=1}^{\infty} x q^{x-1} \\ &= p q \times \frac{1}{(1-q)^2} \\ &= p q \times \frac{1}{p^2} \\ &= \frac{q}{p} \end{aligned}$$

- Variance:  $V(X) = \frac{q}{p^2}$  (prove this)

# Example

- Consider tossing a fair coin until a head appears. Find the probability of getting 5 tails prior to getting the first head [Ans:  $\left(\frac{1}{2}\right)^6$ ].

# Poisson Distribution

- A discrete r.v.  $X$  is said to have a Poisson distribution if its probability function is given by

$$f(x) = \frac{e^{-\lambda} \lambda^x}{x!}; x = 0, 1, 2, \dots, \text{ and } \lambda > 0$$

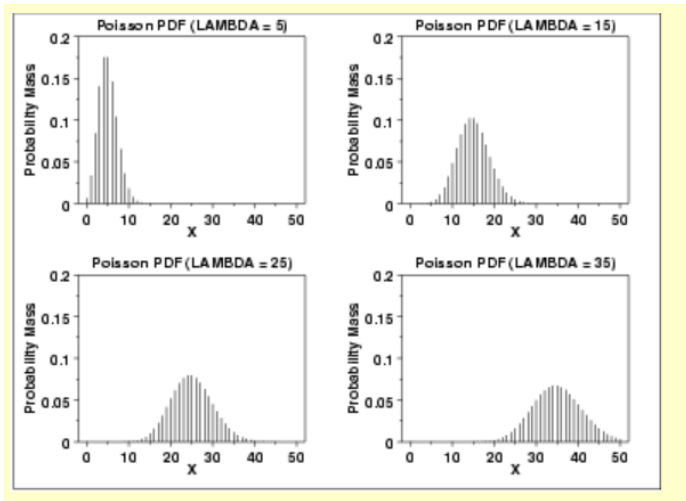
- Here,  $\lambda > 0$  is the parameter of the distribution
- Usually rare events follow a Poisson distribution (for example, number of accidents in 24 hours)
- **Examples:** number of accidents in a day, disintegration in particles, number of incidences during an interval of time, number of typing errors per page

# Poisson Distribution

- Poisson pmf is a valid pmf:

$$\begin{aligned}\sum_{x=0}^{\infty} f(x) &= \sum_{x=0}^{\infty} \frac{e^{-\lambda} \lambda^x}{x!} \\ &= e^{-\lambda} \left[ 1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \dots \right] \\ &= e^{-\lambda} \times e^{\lambda} \\ &= 1\end{aligned}$$

# Shape of a Poisson Distribution



# Mean and Variance of a Poisson R.V.

- Show that for a Poisson r.v.  $X$ ,  $E(X) = V(X) = \lambda$
- **Note:** Poisson distribution is the only distribution in statistics for which mean and variance are equal.

# Example

- (1) A bank finds 5 bad cheques on a single day. Find the probability of finding 10 bad cheques on two consecutive days. (Hint:  $\lambda = 2 \times 5 = 10$ )
- (2) Let  $X$  be a Poisson r.v. such that  $P(X = 1) = P(X = 2)$ . Then  $P(X = 3) = ?$

## Exercise 1 (Poisson)

Tubert-Bitter et al. (A-9) found that the number of serious gastrointestinal reactions reported to the British Committee on Safety of Medicine was 538 for 9,160,000 prescriptions of the anti-inflammatory drug piroxicam. This corresponds to a rate of .058 gastrointestinal reactions per 1000 prescriptions written. Using a Poisson model for probability, with  $\lambda = 0.06$ , find the probability of

- (a) Exactly one gastrointestinal reaction in 1000 prescriptions
- (b) Exactly two gastrointestinal reactions in 1000 prescriptions
- (c) No gastrointestinal reactions in 1000 prescriptions
- (d) At least one gastrointestinal reaction in 1000 prescriptions

# Solution to Exercise 1 (Poisson)

Here,  $\lambda = 0.06$

(a)  $P(X = 1) = P(X \leq 1) - P(X = 0)$

(b)  $P(X = 2) = P(X \leq 2) - P(X \leq 1)$

(c)  $P(X = 0)$

(d)  $P(X \geq 1) = 1 - P(X < 1) = 1 - P(X = 0)$

## Exercise 2 (Poisson)

In a study of the effectiveness of an insecticide against a certain insect, a large area of land was sprayed. Later the area was examined for live insects by randomly selecting squares and counting the number of live insects per square. Experience has shown the average number of live insects per square after spraying to be .5. If the number of live insects per square follows a Poisson distribution, find the probability that a selected square will contain

- (a) Exactly one live insect
- (b) No live insects
- (c) Exactly four live insects
- (d) One or more live insects

## Exercise 3 (Poisson)

In a certain population, an average of 13 new cases of esophageal cancer are diagnosed each year. If the annual incidence of esophageal cancer follows a Poisson distribution, find the probability that in a given year the number of newly diagnosed cases of esophageal cancer will be:

- (a) Exactly 10
- (b) At least eight
- (c) No more than 12
- (d) Between nine and 15, inclusive
- (e) Fewer than seven

## Rapid fire questions (Poisson)

- (1) A bank finds 5 bad cheques on a day. What is the probability of finding 10 bad cheques two consecutive days?
- (2) A typewriter makes 2 errors per page. What is the probability of getting no mistakes if he types 10 pages?
- (3) If 250 litres of water are known to be polluted with  $10^6$  bacteria, what is the probability that a sample of 1cc of the water contains no bacteria? (1 liter = 1000cc)
- (4) Large sheets of metal have faults in random positions but on average have 1 fault per  $10\text{ m}^2$ . What is the probability that a sheet  $5\text{m} \times 8\text{m}$  will have at most one fault?

# Thank You